Special Issue

Symmetry in Uncertainty and Intelligent Decision-Making

Message from the Guest Editor

Symmetry is a fundamental principle in many frameworks for representing uncertainty and supporting intelligent decision-making. It appears in aggregation operators through anonymity and reciprocity axioms, in symmetric divergences such as Jensen–Shannon and Hellinger, and in balanced models based on fuzzy, intuitionistic, neutrosophic, plithogenic, and bipolar sets. Symmetry often improves mathematical tractability and interpretability, while symmetry breaking—asymmetric preferences, heavy-tailed uncertainty, or non-reciprocal trust—can reveal key features of decision problems. This Special Issue invites original contributions on the role of symmetry and asymmetry in modeling, measuring, and reducing uncertainty in intelligent decision-making. Topics include aggregation operators, divergence measures, preference relations, and advances in fuzzy, neutrosophic, plithogenic, rough, and probabilistic frameworks. Applications in multi-criteria decision-making, group consensus, machine learning, large language model evaluation, risk management, healthcare, energy, sustainability, and education are welcome. Symmetry-aware and symmetry-breaking approaches are especially encouraged.

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